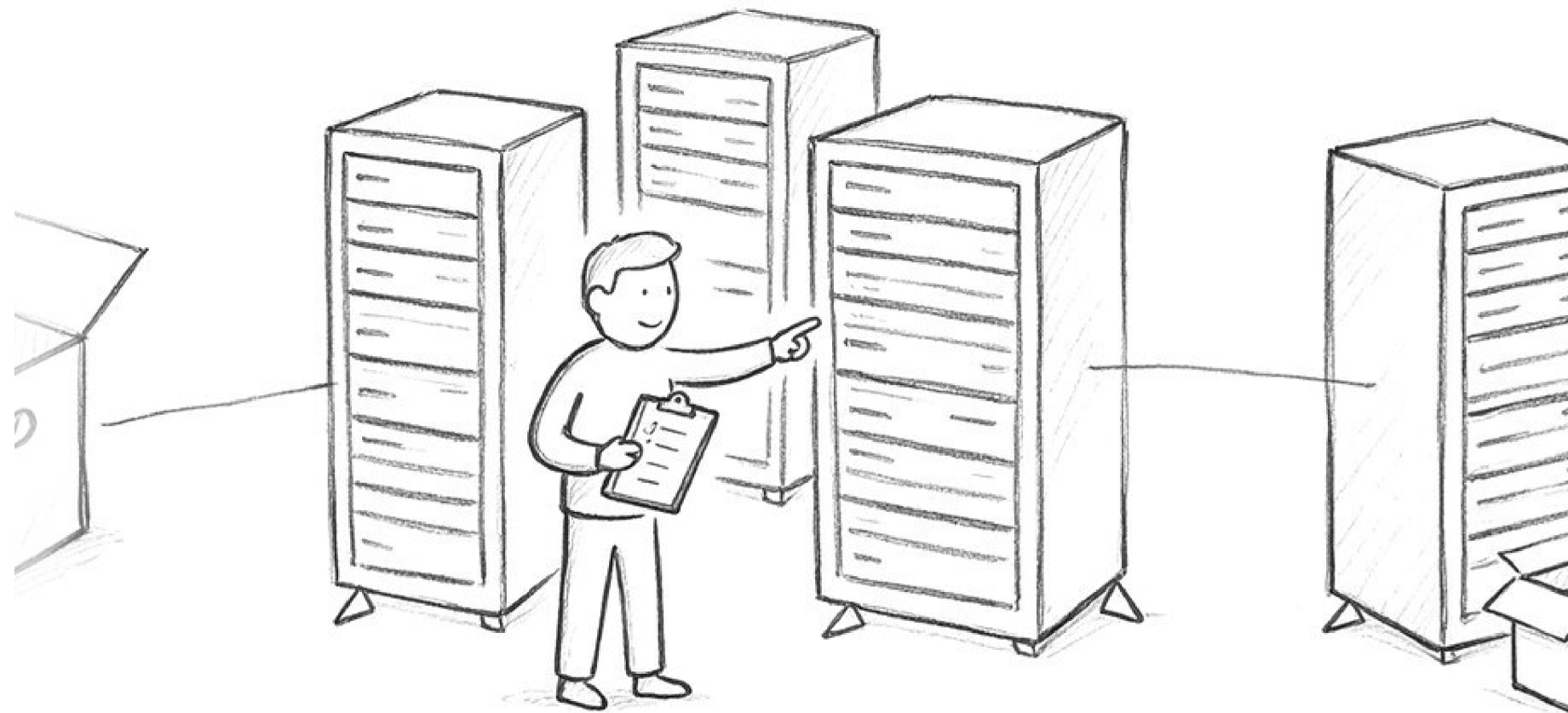


MODULE 4 • KUBERNETES INTERNALS

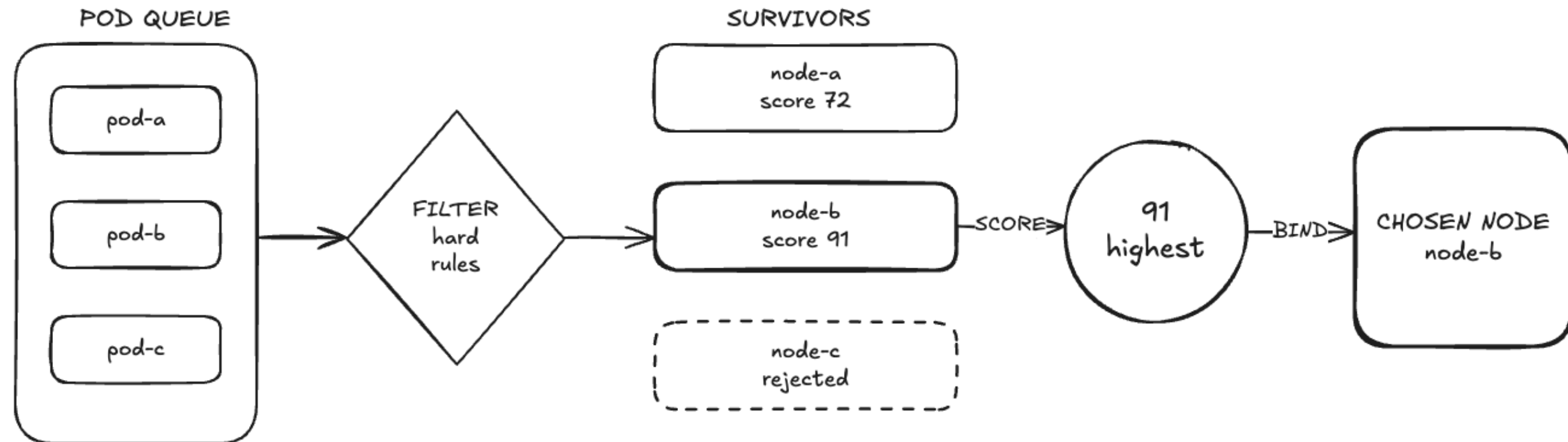
Scheduling and Controllers

How Kubernetes chooses a node and keeps the workload moving toward the desired state.



How a Scheduling Cycle Processes a Pod

Queue → filter → score → bind



Filter: keep valid nodes. Score: rank survivors.

Filter

Remove nodes that fail a hard requirement.

Score

Rank only the nodes that survived filtering.

Bind

Write the selected node to `spec.nodeName`.

Diagnosing a Pending Pod

Read the FailedScheduling event, then test the rule behind each reason.

```
$ kubectl describe pod too-big -n m4-lab
```

Events:

Type	Reason	From
Warning	FailedScheduling	default-scheduler

0/5 nodes are available: 4 Insufficient cpu, 1 intolerated taint.
preemption: 5 Preemption is not helpful

Insufficient cpu

Resources

requests vs allocatable

didn't match selector

Labels

selector vs live labels

intolerated taint

Policy

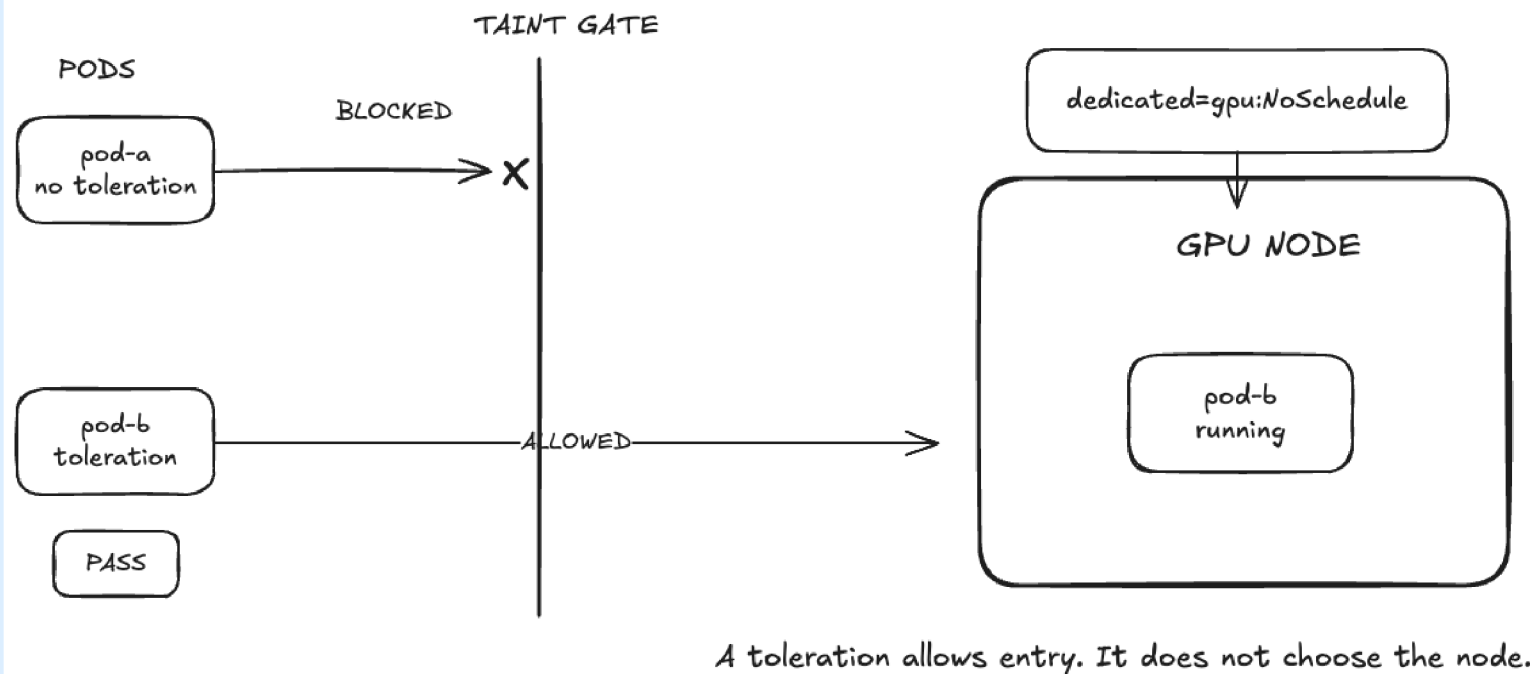
taints vs tolerations

preemption not helpful

Constraints

removing Pods still cannot satisfy
every filter

Taints, Effects, and Tolerations



- 1 A taint is attached to a **node**.
- 2 A toleration is attached to a **Pod**.
- 3 A toleration **permits** placement; it does not choose the node.
- 4 NoExecute can evict Pods already running on the node.

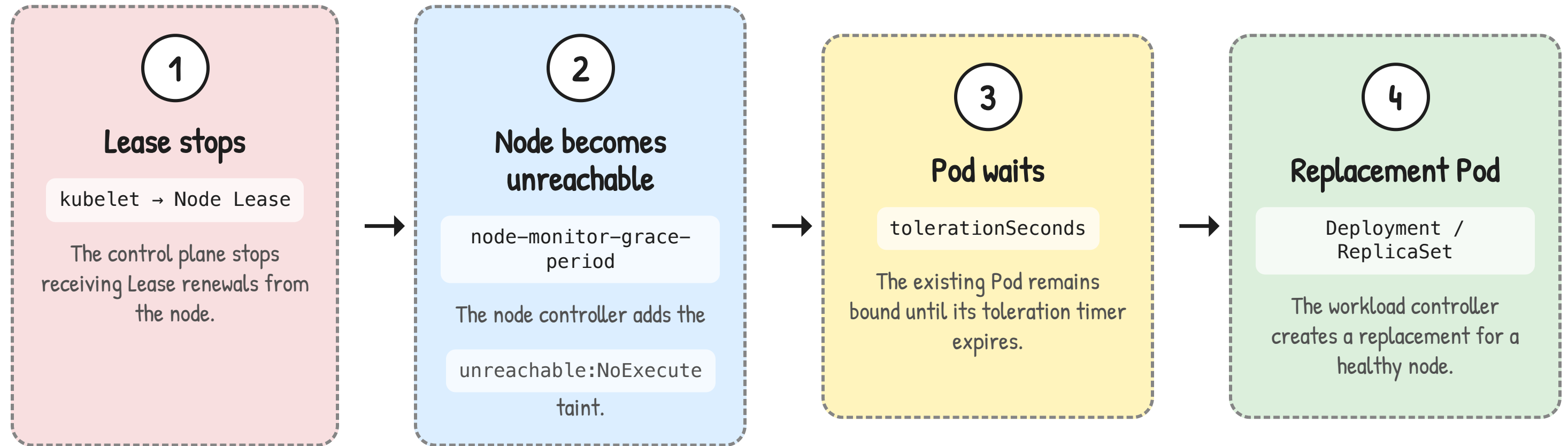
```
dedicated=gpu:NoSchedule
```

```
tolerations:
```

- key: dedicated
- value: gpu
- effect: NoSchedule

How Kubernetes Handles an Unreachable Node

Detection → NoExecute eviction → replacement on a healthy node



Lease stops → unreachable:NoExecute → tolerationSeconds expires → replacement

Scheduling and Controllers

- 1 **Filter** removes invalid nodes; **score** ranks only the survivors.
- 2 **Bind** records the selected node in `spec.nodeName`.
- 3 **FailedScheduling** events identify the rules that rejected the Pod.
- 4 A toleration permits placement; it does not attract a Pod to the node.
- 5 **NoExecute** eviction uses a node-detection clock and a Pod toleration clock.

Next: observe these decisions in the Module 4 lab.